From Ratings to Insights: Airline Satisfaction Study

Objective of the Project

To transform raw airline passenger satisfaction survey data into a structured **data model** in Excel, enabling interactive dashboards and deeper analysis. The goal is to separate facts (measurable events) from dimensions (descriptive categories) so that slicers, PivotTables, and charts can be connected seamlessly.

Data Modeling Framework

1. Fact Table: `fact_airline`

- Contains measurable, numeric data (the "events").
- Example columns:
  - Passenger_ID (primary key)
  - Flight_Distance
  - Departure_Delay
  - Arrival_Delay
  - Satisfaction_Rating (foreign key to `rating_dim`)
  - Class_ID (foreign key to `class_dim`)
  - Customer_Type_ID (foreign key to `cus_type_dim`)
  - Travel_ID (foreign key to `travel_dim`)

Dimension Tables

Dimensions describe attributes that provide context to the facts.

- `class_dim`
  - Class_ID
  - Class_Name (Economy, Business, Economy Plus)
- `rating_dim`
  - Rating_ID
  - Rating_Label (Satisfied, Dissatisfied)
  - Numeric_Score (e.g., 1–5 scale)
- `cus_type_dim`
  - Customer_Type_ID
  - Customer_Type (First-time, Returning)
- `travel_dim`
  - Travel_ID
  - Travel_Type (Business, Personal)

Connections in Data Model

- In Excel's **Power Pivot / Data Model**, link:
  - `fact_airline.Class_ID` → `class_dim.Class_ID`
  - `fact_airline.Rating_ID` → `rating_dim.Rating_ID`
  - `fact_airline.Customer_Type_ID` → `cus_type_dim.Customer_Type_ID`

    - `fact_airline.Travel_ID` → `travel_dim.Travel_ID`
- This creates a **star schema**: one central fact table surrounded by dimension tables

Descriptive Analysis

☞ *What happened?*

- Summarizes the data and shows patterns.
- Example from your dashboard:
    - Business and personal travelers rated food, cleanliness, and seat comfort highly.
    - In-flight entertainment, Wi-Fi, and flight service had more satisfied ratings.
    - Seniors and young passengers showed higher dissatisfaction compared to adults and kids.

2. Diagnostic Analysis

☞ *Why did it happen?*

- Explains reasons behind satisfaction or dissatisfaction.
- Example from your findings:
    - Economy class passengers were more dissatisfied due to delays.
    - First-time travelers had higher dissatisfaction with baggage handling.
    - Male passengers showed greater dissatisfaction across services compared to females.

3. Predictive Analysis

☞ *What is likely to happen?*

- Uses trends to forecast future outcomes.
- Example you could infer:
    - If delays continue, economy passengers will remain the least satisfied group.
    - Long-distance travelers are more likely to give higher satisfaction ratings, so improving long-haul services could boost overall ratings.
    - Online boarding satisfaction suggests digital service improvements will continue to raise ratings.

4. Prescriptive Analysis

☞ *What should we do?*

- Provides recommendations for action.
- Example recommendations:
    - Reduce baggage handling issues for first-time travelers to improve their experience.
    - Focus on minimizing departure and arrival delays, especially in economy class.

- Enhance food and cleanliness standards further, since they strongly influence satisfaction.
- Target service improvements for seniors and younger passengers, who show higher dissatisfaction.

Conclusion

The project converted raw airline passenger satisfaction data into a structured Excel data model with fact and dimension tables. Using this model, interactive dashboards were created to analyze satisfaction across categories such as class, customer type, travel type, and ratings. The analysis highlighted key patterns of satisfaction and dissatisfaction, explained possible reasons, and suggested improvements. Overall, the project shows how data modeling and visualization in Excel can turn raw survey data into clear insights that help airlines improve passenger experience.